

Computer Security

Definition

Computer Security

Properties of a computer system must hold in the presence of a resourced strategic adversary.

The three main security properties

Definition

Confidentiality

Prevention of unauthorized disclosure of information.

Example

The adversary should not be able to read my bank statement.

Definition

Integrity

Prevention of unauthorized modification of information.

Example

The adversary should not be able to change my bank balance.

Definition

Availability

Prevention of unauthorized denial of service.

Example

The adversary should not prevent me from accessing my bank account.

Security Policy

Definition

Security Policy

A high-level description of the security properties that must hold in the system in relation to assets and principals.

Definition

Assets (Objects)

Anything with value that needs to be protected.

Example

Examples of assets :

- data;
- files;
- memory;
- sensitive information.

Definition

Principals (Subjects)

People, computer programs, services, etc.
Principals may not contain the adversary.

Threat Model

Definition

Threat Model

Describes the resources available to the adversary and the adversary's capabilities.

Typical adversary capabilities

- observe;
- influence;
- corrupt;
- exploit available resources.

Definition

Adversary

The adversary is a malicious entity aiming at breaching the security policy.

The adversary is strategic : the adversary chooses the optimal way to use their resources to mount an attack that violates the security properties.

Vulnerabilities and Threats

Definition

Vulnerability

A specific weakness that could be exploited by adversaries with interest in different assets.

Example

Examples :

- The banking API is not protected.
- The password appears in plain text on my screen.

Definition

Threat

The feared event or the goal of the adversary that we do not want to materialize.

Example

Examples :

- A hacker wants to retrieve money by breaking into the bank's system.
- A student wants to learn my password by looking over my shoulder.

Definition

Harm

The bad thing that happens when the threat materializes.

Example

Examples :

- The adversary steals money.
- The adversary blocks access to the bank.
- The adversary learns my password.
- The adversary reads a message.

Relationship Between Concepts**Key idea**

If the security properties hold within the threat model, it means that :

- there are no vulnerabilities that can be exploited;
- threats cannot materialize;
- no harm happens.

Security Mechanisms**Definition****Security Mechanism**

A technical mechanism used to ensure that the security policy is not violated by an adversary within the threat model.

Key idea

Security mechanisms protect the system against adversaries whose resources and capabilities are described by the threat model.

When is a System Secure ?**Definition****Secure System**

A system is secure if an adversary constrained by a specific threat model cannot violate the security policy.

Summary

A secure system depends on :

- a clearly defined security policy;
- identified assets and principals;
- a realistic threat model;
- effective security mechanisms.

Security Argument**Definition****Security Argument**

A rigorous argument that the security mechanisms in place are indeed effective in maintaining the security policy.

The argument can be verbal or mathematical and is subject to the assumptions of the threat model.